

# No Finite Window Forces Banach-Valued Fekete Convergence

Automated Math Discovery Candidate Counterexample

August 11, 2026

**Status.** Candidate full negative answer to Question 3.2 of Kulikov–Shao, arXiv:2411.17380v1, likely valid pending expert review. For every proposed finite window function, the required inequalities admit a nonconvergent sequence already in the Euclidean plane.

## Source Question

Kulikov and Shao prove a Banach-valued Fekete lemma when  $\|v_{n+m}\| \leq \|v_n + v_m\|$  is imposed for all pairs [1]. After constructing planar counterexamples for several wide but prescribed windows, they ask on page 8:

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exist functions  $f(n)$  such that the Banach space-valued version of the Fekete’s subadditivity lemma is true if we only assume it for  $n \leq m \leq f(n)$ .

**Question 3.2.** *Let  $X$  be a uniformly convex Banach space. Does there exist a function  $f : \mathbb{N} \rightarrow \mathbb{N}$  such that for all sequences  $v_n \in X$  with  $\|v_{n+m}\| \leq \|v_n + v_m\|$  for  $n \leq m \leq f(n)$  the limit  $\lim_{n \rightarrow \infty} \frac{v_n}{n}$  exists?*

By the Proposition [3.1](#) if  $X$  is one-dimensional then  $f(n) = 2n$  is enough, but already for the two-dimensional Hilbert spaces  $X$  we need a superlinear function  $f$ .

In other words, for a fixed uniformly convex space  $X$ , can any function  $f : \mathbb{N} \rightarrow \mathbb{N}$  make the restricted inequalities

$$\|v_{n+m}\| \leq \|v_n + v_m\| \quad (n \leq m \leq f(n))$$

force convergence of  $v_n/n$ ? The next theorem answers no for  $X = \mathbb{R}^2$ , hence answers the question negatively.

## Answer and Proof Intuition

**Theorem 1** (No finite window in the Euclidean plane). *For every function  $f : \mathbb{N} \rightarrow \mathbb{N}$ , there is a sequence  $(v_n)_{n \geq 1} \subset \mathbb{R}^2$  such that*

$$\|v_{n+m}\|_2 \leq \|v_n + v_m\|_2 \quad \text{whenever } n \leq m \leq f(n),$$

*but  $(v_n/n)$  does not converge.*

The construction uses two scales. First, partition  $\mathbb{N}$  into finite blocks chosen recursively from  $f$ , so that an allowed pair can meet at most two consecutive blocks. Second, rotate the direction by  $c/k$  between blocks and add radial slack proportional to the tail  $\sum_{j \geq k} j^{-2}$ . The angular increments have divergent sum, so the directions rotate forever; their squares have convergent sum, so a vanishing amount of radial slack can pay for the Euclidean cosine loss.

## Proof

Fix  $f : \mathbb{N} \rightarrow \mathbb{N}$ . Put  $N_1 = 1$ ,  $N_2 = 2$ , and recursively choose  $N_{k+2}$  so large that

$$N_{k+2} > N_{k+1} \quad \text{and} \quad N_{k+2} > \max\{f(n) : 1 \leq n < N_{k+1}\}. \quad (1)$$

This is possible because the set in the maximum is finite. Let

$$B_k = \{n \in \mathbb{N} : N_k \leq n < N_{k+1}\}.$$

Every block is nonempty. More importantly, if  $n \in B_k$  and  $n \leq m \leq f(n)$ , then (1) gives

$$m < N_{k+2}, \quad m \in B_k \cup B_{k+1}. \quad (2)$$

Set  $c = 1/4$ , and for  $k \geq 1$  define

$$\alpha_1 = 0, \quad \alpha_{k+1} = \alpha_k + \frac{c}{k}, \quad s_k = 2c^2 \sum_{j=k}^{\infty} \frac{1}{j^2}. \quad (3)$$

Then  $s_k \downarrow 0$ ,

$$s_k - s_{k+1} = \frac{2c^2}{k^2}, \quad (4)$$

and

$$s_1 = \frac{\pi^2}{48} < \sqrt{2} - 1, \quad (1 + s_1)^2 < 2. \quad (5)$$

For  $n \in B_k$ , put

$$r_n = n(1 + s_k), \quad v_n = r_n(\cos \alpha_k, \sin \alpha_k). \quad (6)$$

We verify the restricted Fekete inequalities. Take  $n \in B_k$  and  $n \leq m \leq f(n)$ . By (2), there are only two cases.

If  $m \in B_k$ , the vectors  $v_n, v_m$  point in the same direction. If  $n + m \in B_h$ , then  $h \geq k$ , so  $s_h \leq s_k$ , and therefore

$$r_{n+m} = (n + m)(1 + s_h) \leq (n + m)(1 + s_k) = r_n + r_m = \|v_n + v_m\|_2.$$

Suppose instead that  $m \in B_{k+1}$ . Now  $n + m \in B_h$  for some  $h \geq k + 1$ , so  $s_h \leq s_{k+1}$ . Consequently, with  $D = r_n + r_m - r_{n+m}$ , equations (4) and (6) give

$$D \geq n(s_k - s_{k+1}) = \frac{2c^2 n}{k^2}. \quad (7)$$

Hence

$$(r_n + r_m)^2 - r_{n+m}^2 = D(r_n + r_m + r_{n+m}) \geq \frac{2c^2 nm}{k^2}, \quad (8)$$

because  $r_n + r_m + r_{n+m} \geq r_m \geq m$ .

The angle between  $v_n$  and  $v_m$  is  $c/k$ . Using  $2(1 - \cos x) \leq x^2$ ,  $s_k, s_{k+1} \leq s_1$ , and (5), the angular loss is bounded by

$$\begin{aligned} 2r_n r_m \left(1 - \cos \frac{c}{k}\right) &\leq r_n r_m \frac{c^2}{k^2} \\ &\leq (1 + s_1)^2 n m \frac{c^2}{k^2} < \frac{2c^2 n m}{k^2}. \end{aligned} \tag{9}$$

Combining (8) and (9),

$$\begin{aligned} \|v_n + v_m\|_2^2 - r_{n+m}^2 &= (r_n + r_m)^2 - r_{n+m}^2 - 2r_n r_m \left(1 - \cos \frac{c}{k}\right) \\ &\geq 0. \end{aligned}$$

Thus every required inequality holds.

It remains to prove nonconvergence. Along the block beginnings,

$$\frac{v_{N_k}}{N_k} = (1 + s_k)(\cos \alpha_k, \sin \alpha_k), \quad s_k \longrightarrow 0, \quad \alpha_k = c \sum_{j < k} \frac{1}{j}.$$

The angles tend to infinity, while their increments tend to zero. More explicitly, for each  $k$  choose  $\ell(k) > k$  minimal with  $\alpha_{\ell(k)} - \alpha_k \geq \pi$ . The divergent harmonic series makes this possible, and minimality gives

$$0 \leq \alpha_{\ell(k)} - \alpha_k - \pi < \frac{c}{\ell(k) - 1} \longrightarrow 0.$$

Therefore the unit directions at indices  $k$  and  $\ell(k)$  become asymptotically antipodal. The subsequence  $v_{N_k}/N_k$  is not Cauchy, so  $v_n/n$  does not converge. This proves Theorem 1.  $\square$

## Scope, Novelty Check, and Review Focus

The theorem is stronger than a counterexample to one proposed growth rate: the block boundaries are constructed after seeing an arbitrary function  $f$ . It also uses the two-dimensional Hilbert space, so no pathology of an infinite-dimensional or merely strictly convex Banach space is involved.

A bounded novelty check through August 11, 2026 searched the exact arXiv id and title, the wording of Question 3.2, and combinations of “Banach-valued Fekete lemma,” “window function,” and “uniformly convex.” The current arXiv record remains v1, and no later primary source resolving the question was found. The construction and square-summable/nonsummable device should be checked by a specialist, with particular attention to the locality implication (2) and the budget comparison (8)–(9).

## References

- [1] A. Kulikov and F. Shao, *Fekete’s lemma in Banach spaces*, arXiv:2411.17380v1 (2024), Question 3.2, p. 8.